

# Conformal Inference Assignment 1

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Wen, Zehai

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**Remark** All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

## Exercise 1

Consider split conformal prediction with  $\mathcal{Y} = \mathbb{R}$  and a given model  $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}$ . Construct a score function  $s(x, y)$  that would yield intervals of the form  $\mathcal{C}(x) = [\hat{f}(x) - \gamma, \hat{f}(x) + 2\gamma]$  for some  $\gamma > 0$ .

**Solution** Put

$$s(x, y) = \begin{cases} 2|\hat{f}(x) - y| & y < \hat{f}(x) \\ |\hat{f}(x) - y| & y \geq \hat{f}(x). \end{cases}$$

Then, with  $\hat{q}$  being a quantile of the scores on the validation set and  $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\}$ , we will have the desired interval with  $\gamma = \frac{\hat{q}}{2}$ . ///

## Exercise 2

Consider split conformal prediction with  $\mathcal{Y} = \mathbb{R}^d$  and a given model  $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}^d$  whose coordinates are  $\hat{f}(x) = (\hat{f}_1(x), \dots, \hat{f}_d(x))$ . Construct a score function  $s(x, y)$  that would yield intervals of the form

$$\mathcal{C}(x) = \prod_{j=1}^d [\hat{f}_j(x) - c, \hat{f}_j(x) + c]$$

for a  $c > 0$ .

**Solution** Put

$$s(x, y) = \max_{j=1, \dots, d} |\hat{f}_j(x) - y_j|.$$

Then, with  $\hat{q}$  being a quantile of the scores on the validation set and  $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\}$ , we will have the desired interval with  $c = \hat{q}$ . ///

## Exercise 3

Consider split conformal prediction with  $\mathcal{Y} = \mathbb{R}$ , two given models  $\hat{f}_i : \mathcal{X} \rightarrow \mathbb{R}$  for  $i = 1, 2$ , and the score function

$$s(x, y) = \min_{i=1,2} |y - \hat{f}_i(x)|.$$

Derive an explicit expression for  $\mathcal{C}(x)$  in terms of  $\hat{f}_1(x)$ ,  $\hat{f}_2(x)$ , and the quantile  $\hat{q}$  of the scores on the validation set. What is the shape of the prediction when  $\hat{q}$  is small or when it is large? Briefly explain why this might be attractive if the conditional distribution of  $Y$  given  $X = x$  is bimodal.

**Solution** By definition, it is immediately clear that

$$\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\} = [\hat{f}_1(x) - \hat{q}, \hat{f}_1(x) + \hat{q}] \cup [\hat{f}_2(x) - \hat{q}, \hat{f}_2(x) + \hat{q}].$$

If  $\hat{f}_1(x) = \hat{f}_2(x)$ , then  $\mathcal{C}(x)$  is an interval of length  $2\hat{q}$  centered at  $\hat{f}_1(x) = \hat{f}_2(x)$ . If we assume that the two models are different at  $x$ , that is,  $\hat{f}_1(x) \neq \hat{f}_2(x)$ , then  $\mathcal{C}(x)$  is the union of two intervals when  $\hat{q}$  is small enough, and it is a single interval when  $\hat{q}$  is large enough. If the conditional distribution of  $Y$  given  $X = x$  is bimodal, then the score might be of interest because  $\hat{f}_1(x)$  and  $\hat{f}_2(x)$  might capture the two modes of the distribution and the predicted interval will be centred at both modes. ////

## Exercise 4

Fix  $n > 2$ . Construct a random vector  $(X_1, \dots, X_n)$  defined on some probability space such that  $(X_i, X_j)$  is exchangeable for all  $i \neq j$  but  $(X_1, \dots, X_n)$  is not exchangeable.

**Solution** Let  $(X, Y)$  be a jointly normal random vector. It is exchangeable if and only if  $\mathbb{E}[X] = \mathbb{E}[Y]$  and  $\text{Var}(X) = \text{Var}(Y)$ . With this in mind, let  $(X_1, \dots, X_n)$  be a jointly normal random vector such that  $\mathbb{E}[X_i] = \mathbb{E}[X_j]$  and  $\text{Var}(X_i) = \text{Var}(X_j)$  for all  $i \neq j$  but  $\text{Cov}(X_1, X_2) \neq \text{Cov}(X_1, X_3)$ . Then  $(X_i, X_j)$  is exchangeable for all  $i \neq j$  but  $(X_1, \dots, X_n)$  is not exchangeable. For a concrete example, say  $\mathbb{E}[X_i] = 0$  and  $\text{Var}(X_i) = 1$  for all  $i$ ,  $\text{Cov}(X_1, X_2) = -1$ , and all other covariances are zero. ////

## Exercise 5

Fix  $n \geq 1$ ,  $\tau \in (0, 1)$ . For any vector  $v \in \mathbb{R}^n$ , define  $\text{Quantile}(v; \tau) = v_{(\lceil \tau n \rceil)}$ , where  $v_{(1)} \leq v_{(2)} \leq \dots \leq v_{(n)}$  are the order statistics of  $v$ . Under what condition is it true that, for all  $v \in \mathbb{R}^n$ ,  $\text{Quantile}(v; 1 - \tau) = -\text{Quantile}(-v; \tau)$ ?

**Solution** The order statistics of  $-v$  are given by  $(-v)_{(i)} = -v_{(n-i+1)}$  for  $i = 1, \dots, n$ . Therefore,  $\text{Quantile}(v; 1 - \tau) = -\text{Quantile}(-v; \tau)$  holds for a vector  $v \in \mathbb{R}^n$  if and only if

$$v_{(\lceil (1-\tau)n \rceil)} = -((-v)_{(\lceil \tau n \rceil)}) = -(-v_{(n-\lceil \tau n \rceil+1)}) = v_{(n-\lceil \tau n \rceil+1)}.$$

To make this work for all  $v \in \mathbb{R}^n$ , the condition we must pose is that

$$\lceil (1 - \tau)n \rceil = n - \lceil \tau n \rceil + 1.$$

This can happen for instance when  $\tau = 1/2$  and  $n$  is odd. ////

## Exercise 6

For  $z \in \mathbb{R}^n$ , write  $\hat{F}_z(v) = \frac{1}{n} \sum_{i=1}^n \chi_{\{z_i \leq v\}}$  for the empirical cumulative distribution function of  $z$  with  $\hat{F}_z(-\infty) = 0$  and  $\hat{F}_z(+\infty) = 1$ . Write

$$\text{Quantile}(z; \tau) = \inf \left\{ v \in \mathbb{R} \mid \hat{F}_z(v) \geq \tau \right\}$$

for the empirical  $\tau$ -quantile of  $z$  for any  $\tau \in [0, 1]$ . Consequently,  $\text{Quantile}(z; 0) = -\infty$ .

Prove that, for any  $z \in \mathbb{R}^n$ , the following statements are true:

1.  $\hat{F}_z(v) = \sup \{ \tau \in [0, 1] \mid \text{Quantile}(z; \tau) \leq v \}$  for all  $v \in \mathbb{R}$ .

2.  $\text{Quantile}(z; \hat{F}_z(v)) \leq v$  for all  $v \in \mathbb{R}$ .
3.  $\hat{F}_z(\text{Quantile}(z; \tau)) \geq \tau$  for all  $\tau \in [0, 1]$ .
4. If  $z$  has no repeated values, then  $\hat{F}_z(\text{Quantile}(z; \tau)) = \frac{\lfloor n\tau \rfloor}{n}$  for all  $\tau \in [0, 1]$ .

**Solution** Let  $z \in \mathbb{R}^n$  be given and fixed. Let us introduce a notation  $S_\tau$  for the set of all  $v \in \mathbb{R}$  such that  $\hat{F}_z(v) \geq \tau$ . Then, by definition,  $\text{Quantile}(z; \tau) = \inf S_\tau$ . All four statements depend on the following lemma:

**Lemma 1.** If  $v \in \mathbb{R}$  and  $\tau \in [0, 1]$ , then  $\text{Quantile}(z; \tau) \leq v$  if and only if  $\tau \leq \hat{F}_z(v)$ .

*Proof.* If  $\tau \leq \hat{F}_z(v)$ , then  $v \in S_\tau$  and  $v \geq \inf S_\tau = \text{Quantile}(z; \tau)$ .

Let us focus on the converse. Suppose that  $\text{Quantile}(z; \tau) \leq v$ . For simplicity, put  $q = \text{Quantile}(z; \tau)$ . If  $\tau = 0$ , then  $\hat{F}_z(v) \geq 0 = \tau$  holds trivially. Assume from now on that  $\tau > 0$ . Then  $q$  is finite number. Choose  $\delta > 0$  small enough such that no points  $z_i$  lies between  $q$  and  $q + \delta$ , which is always doable because  $z$  has only finitely many points. Since  $q$  is the infimum, choose  $s \in S_\tau$  such that  $q \leq s < q + \delta$ . Then, we arrive at

$$\hat{F}_z(v) \geq \hat{F}_z(q) = \hat{F}_z(s) \geq \tau$$

by applying in the first inequality the monotonicity of  $\hat{F}_z$  and in the second equality the fact that no points  $z_i$  lies between  $q$  and  $q + \delta$ . The lemma is proved.  $\square$

We now prove the four statements. To prove the first statement, let  $v \in \mathbb{R}$  be given and we have, by the lemma:

$$\{\tau \in [0, 1] \mid \text{Quantile}(z; \tau) \leq v\} = \{\tau \in [0, 1] \mid \tau \leq \hat{F}_z(v)\} = [0, \hat{F}_z(v)]$$

and the supremum of the interval is just  $\hat{F}_z(v)$ . The second and the third statements are already included in the lemma. The fourth statement is trivial given the special assumption that  $z$  has no repeated values.////

## Exercise 7

Consider the conformal prediction with  $\mathcal{Y} = \mathbb{R}$  with a given model  $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$  and calibration points  $(X_1, Y_1), \dots, (X_n, Y_n)$ . Put

$$\hat{q}_L = \text{Quantile} \left( \left\{ \hat{f}(X_i) - Y_i \right\}_{i=1}^n ; \left(1 - \frac{\alpha}{2}\right) \left(1 + \frac{1}{n}\right) \right),$$

$$\hat{q}_R = \text{Quantile} \left( \left\{ Y_i - \hat{f}(X_i) \right\}_{i=1}^n ; \left(1 - \frac{\alpha}{2}\right) \left(1 + \frac{1}{n}\right) \right),$$

and

$$\mathcal{C}(X_{n+1}) = \left[ \hat{f}(X_{n+1}) - \hat{q}_L, \hat{f}(X_{n+1}) + \hat{q}_R \right].$$

Prove that

$$P \{Y_{n+1} \in \mathcal{C}(X_{n+1})\} \geq 1 - \alpha$$

by showing both

$$P \left\{ Y_{n+1} < \hat{f}(X_{n+1}) - \hat{q}_L \right\} \leq \frac{\alpha}{2}$$

and

$$P \left\{ Y_{n+1} > \hat{f}(X_{n+1}) + \hat{q}_R \right\} \leq \frac{\alpha}{2},$$

assuming that  $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$  is exchangeable.

**Solution** We have

$$P \left\{ Y_{n+1} < \hat{f}(X_{n+1}) - \hat{q}_L \right\} = P \left\{ \hat{f}(X_{n+1}) - Y_{n+1} > \hat{q}_L \right\} = 1 - P \left\{ \hat{f}(X_{n+1}) - Y_{n+1} \leq \hat{q}_L \right\} \leq 1 - \left( 1 - \frac{\alpha}{2} \right) = \frac{\alpha}{2}$$

where the last inequality uses Theorem 3.2 in the textbook with score  $S_i = \hat{f}(X_i) - Y_i$ . This is legitimate because the data is assumed to be exchangeable and the score is trivially symmetric in case of split conformal prediction. The right hand side inequality is proved in exactly the same way but with score  $S_i = Y_i - \hat{f}(X_i)$ . Combine the two, we have the desired marginal coverage. The proof is complete. ///

## Exercise 8

Consider the full conformal prediction with  $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$  being exchangeable and a symmetric score function  $s$ , prove that

$$P \{ Y_{n+1} \in \mathcal{C}(X_{n+1}) \} \leq 1 - \alpha + \mathbb{E} \left[ \frac{1}{n+1} \sum_{i=1}^{n+1} \chi_{\{S_i = \hat{q}\}} \right]$$

where

$$\hat{q} = \text{Quantile}(S_1, \dots, S_{n+1}; 1 - \alpha).$$

Give an example showing that this bound is tighter than

$$P \{ Y_{n+1} \in \mathcal{C}(X_{n+1}) \} \leq \frac{\lceil (1 - \alpha)(n + 1) \rceil}{n + 1} + \epsilon_{\text{tie}}$$

where  $\epsilon_{\text{tie}}$  is the probability of  $S_{n+1}$  being equal to some  $S_j$  for  $j \in \{1, \dots, n\}$ .

**Solution** As already discussed in the lecture and in the textbook,  $Y_{n+1} \in \mathcal{C}(X_{n+1})$  if and only if  $S_{n+1} \leq \hat{q}$  and so

$$P \{ Y_{n+1} \in \mathcal{C}(X_{n+1}) \} = P \{ S_{n+1} \leq \hat{q} \} = P \{ S_{n+1} < \hat{q} \} + P \{ S_{n+1} = \hat{q} \}.$$

Given that  $(S_1, \dots, S_n, S_{n+1})$  is exchangeable,  $P \{ S_{n+1} = \hat{q} \} = P \{ S_i = \hat{q} \}$  for all  $i \in \{1, \dots, n\}$  and

$$P \{ S_{n+1} = \hat{q} \} = \frac{1}{n+1} \sum_{i=1}^{n+1} P \{ S_i = \hat{q} \} = \mathbb{E} \left[ \frac{1}{n+1} \sum_{i=1}^{n+1} \chi_{\{S_i = \hat{q}\}} \right].$$

The only thing left is to show that  $P \{ S_{n+1} < \hat{q} \} \leq 1 - \alpha$ . To do so, first observe that  $S_{n+1} < \hat{q}$  if and only if  $S_{n+1} < S_{(k)}$ , where  $k = \lceil (1 - \alpha)(n + 1) \rceil$ . Thus:

$$P \{ S_{n+1} < \hat{q} \} = P \{ S_{n+1} < S_{(k)} \} \leq \frac{\lceil (1 - \alpha)(n + 1) \rceil - 1}{n + 1} = \frac{\lfloor (1 - \alpha)(n + 1) \rfloor}{n + 1} \leq \frac{(1 - \alpha)(n + 1)}{n + 1} = 1 - \alpha.$$

This completes the proof.

Finally, in case when the random variables  $S_i$ 's are all continuous random variables,  $\epsilon_{\text{tie}} = P \{ S_{n+1} = \hat{q} \} = 0$  and evidently the bound that is worked out in this exercise is better. In case when the random variables  $S_i$ 's are discrete and can only take very limited values, it may happen that  $S_1, \dots, S_n$  has taken all possible values and  $\epsilon_{\text{tie}} = 1$  so that the bound worked out here is really the only available bound. ///

## Exercise 9

Given data set  $(X_1, Y_1), \dots, (X_n, Y_n) \in \mathcal{X} \times \mathcal{Y}$  where  $\mathcal{Y} = \mathbb{R}$ , consider split conformal prediction with two given models  $\hat{f}_i : \mathcal{X} \rightarrow \mathbb{R}$ , the score functions

$$s_i(x, y) = |y - \hat{f}_i(x)|,$$

and two split conformal prediction sets

$$\mathcal{C}_i(X_{n+1}) = \{y \in \mathcal{Y} \mid s_i(X_{n+1}, y) \leq \hat{q}_i\},$$

where

$$\hat{q}_i = \text{quantile} \left\{ \{s_i(X_j, Y_j)\}_{j=1}^n; (1 - \alpha)(1 + \frac{1}{n}) \right\}.$$

Put

$$L_i = \frac{1}{n} \sum_{j=1}^n (Y_j - \hat{f}_i(X_j))^2$$

and  $I = \arg \min_{i \in \{1, 2\}} L_i$ . Explain why

$$P \{Y_{n+1} \in \mathcal{C}_I(X_{n+1})\} \geq 1 - \alpha$$

may not hold even if we assume that  $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$  are exchangeable.

**Solution** In one sentence, the reason is that even if we assume that the three random vectors  $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ ,  $s_1(X_1, Y_1), \dots, s_1(X_{n+1}, Y_{n+1})$ , and  $s_2(X_1, Y_1), \dots, s_2(X_{n+1}, Y_{n+1})$  are all exchangeable, in general it is still not true that  $s_I(X_1, Y_1), \dots, s_I(X_{n+1}, Y_{n+1})$  is exchangeable and thus the theorem guaranteeing the marginal coverage does not in general apply.

To show that things can indeed fail, we give a simple counterexample. Put  $\mathcal{X} = \mathbb{R}^2$ ,  $n = 1$ ,  $\alpha = \frac{1}{2}$ ,  $X_j = (X_j^{(1)}, X_j^{(2)})$ ,  $X_j$  follows  $N_2(0, I_2)$  distribution,  $Y_j = X_j^{(1)} + X_j^{(2)}$ ,  $\hat{f}_1(x) = x^{(1)}$  and  $\hat{f}_2(x) = x^{(2)}$ . It is clear that, for  $i = 1, 2$ ,  $\hat{q}_i = s_i(X_1, Y_1)$ ,  $s_i(X_j, Y_j) = |X_j^{(2-i)}|$ , and  $L_i = (X_1^{(2-i)})^2$ . Thus,

$$\begin{aligned} & P \{s_I(X_2, Y_2) \leq \min(s_1(X_1, Y_1), s_2(X_1, Y_1))\} \\ &= P \{s_1(X_2, Y_2) \leq \min(s_1(X_1, Y_1), s_2(X_1, Y_1)), L_1 < L_2\} \\ &+ P \{s_2(X_2, Y_2) \leq \min(s_1(X_1, Y_1), s_2(X_1, Y_1)), L_1 > L_2\} \\ &= P \left\{ |X_2^{(2)}| \leq |X_1^{(2)}| \leq |X_1^{(1)}| \right\} + P \left\{ |X_2^{(1)}| \leq |X_1^{(1)}| \leq |X_1^{(2)}| \right\} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \end{aligned}$$

and

$$P \{Y_2 \in \mathcal{C}_I(X_2)\} = P \{s_I(X_2, Y_2) \leq \min(s_1(X_1, Y_1), s_2(X_1, Y_1))\} = \frac{1}{3} < \frac{1}{2} = 1 - \alpha.$$

This completes the counterexample. ////